

Interferomatic Manual

Welcome! Interferomatic is the software that we use to get data with our dual frequency comb spectrometer.

It's easy to use once you know the basic steps to get a baseline, collect data, and analyze those data.

Basic Steps

1. First, start Interferomatic by clicking the icon on the desktop, or run these commands:

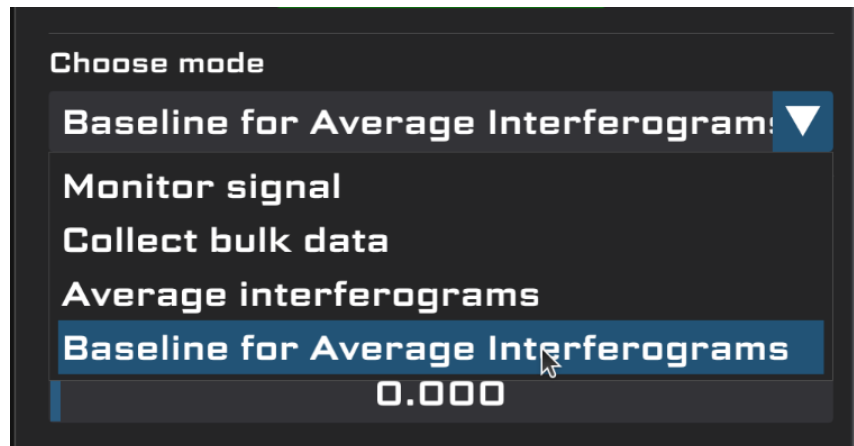
```
cd interferomatic
```

```
python3 main.py
```

2. Now we need to get a baseline. **IMPORTANT!** This is required if you want to get a reading of some material sample and look at its transmission/absorption lines. Without a baseline, you will get a lot of irrelevant junk that makes it harder to see these lines. However, if you want to analyze the incoming spectrum of the laser, then you do not want a baseline and so you can skip this step. This also only applies if you want to collect data by averaging interferograms. If you are monitoring the live signal or collecting bulk data then this baseline step doesn't apply.

To get a baseline,

- click the "Choose mode" dropdown and select "Baseline for Average Interferograms"



- choose a good number of at least 1000 or so for the number of interferograms to collect so that you don't have excess noise. Note that our setup generates approximately 25-50 interferograms per second, so 1000 interferograms will take about half a minute

- choose a good file name to save your baseline to in the "Save file" selector
- change the rest of the settings to match what you'll use later when averaging interferograms
- click the "Start" button to start collecting data and it should now say "Stop"
- wait
- once collection has finished, the "Stop" button should turn back to "Start"
- make sure the baseline is saved to the file location specified

3. Finally, let's collect our data by averaging interferograms.

- click the "Choose mode" dropdown again and select the "Average Interferograms" option
- change the interferograms to collect to be much higher. 100000 (one hundred thousand) is a common value for our experiments
- change the "Save file" location to be a different file than the baseline
- click the "Start" button to start collecting data. It should turn into "Stop" like with getting the baseline
- wait
- once the data collection is done, once again, make sure the data is saved in the specified file